Customer Churn Prediction Using Random Forest: A Data Science Approach

Prepared by: Vijaya Gokul

1. Project Overview :

Customer churn refers to the loss of clients or subscribers over time. Predicting churn is essential for businesses to proactively retain customers, optimize marketing strategies, and improve service offerings.

This project focuses on building a **Random Forest model** to identify customers who are likely to churn based on historical behavior, account information, and usage patterns. Accurate churn prediction enables targeted interventions to reduce customer attrition.

2. Methodology :

Churn prediction is a **binary classification problem**, where the target variable indicates whether a customer will churn (`1`) or remain (`0`).

Model Selection:

The project uses **Random Forest**, an ensemble learning method that constructs multiple decision trees and aggregates their predictions. Random Forest can capture nonlinear relationships, handle feature interactions, and is robust to outliers. Additionally, it provides **feature importance scores**, helping identify which factors most influence churn

Evaluation Metrics :

To evaluate model performance, the following metrics are used:

- **Accuracy:** Measures overall correctness of predictions.

- **Precision:** Indicates the proportion of predicted churners that actually churned.
- **Recall (Sensitivity):** Measures the ability to identify actual churners.
- **F1-score:** Balances precision and recall for imbalanced datasets.
- **ROC-AUC:** Evaluates the model's discrimination ability across thresholds.

3. Feature Engineering:

Key features created from the dataset:

- **Customer Activity Metrics:** Number of logins, purchases, interactions, engagement frequency.
- **Tenure Features:** Account age, time since last activity.
- **Usage Patterns:** Average transaction value, service usage trends, drop in engagement.
- **Demographics and Behavior:** Age, location, subscription type, support tickets.
- **Derived Features:** Ratios of active days/total days, flags for sudden activity drops.

4.Exploratory Data Analysis (EDA) and Insights:

4.1Churn Distribution:

The dataset is imbalanced; fewer churners than retained customers.

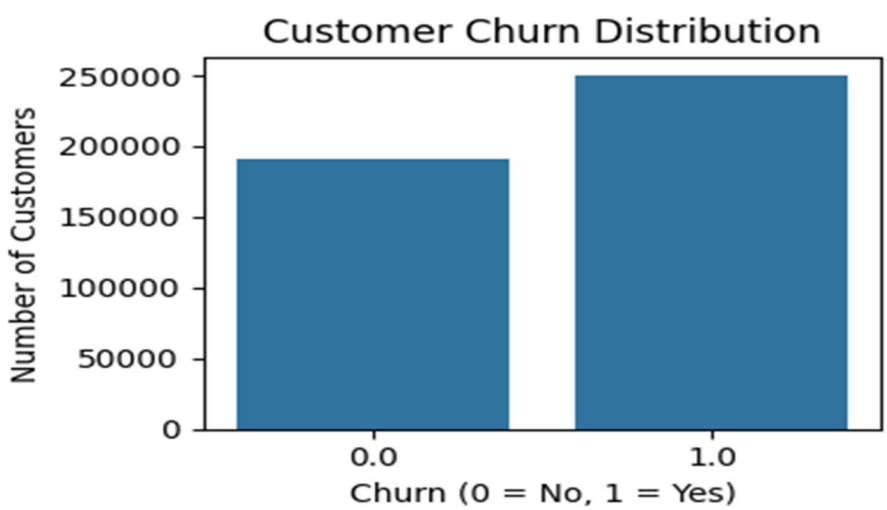

Figure 1. churn distribution

4.2 Top 6 drivers of churn :

This is the most important visual for your "Results" section. It highlights the variables that have the highest predictive power—likely **Support Calls**, **Payment Delay**, and **Tenure**.

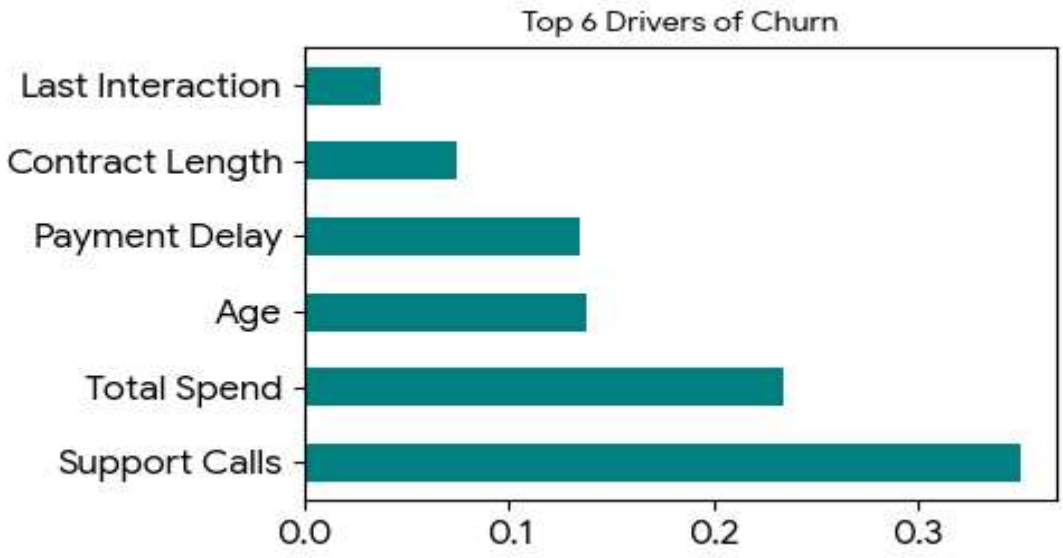

Figure 2 top 6 drivers of churn

4.3 Business Insight: Support Calls vs. Churn

A boxplot that clearly demonstrates the "threshold" where customer frustration leads to churn. This visual directly supports your recommendation for improving customer support.

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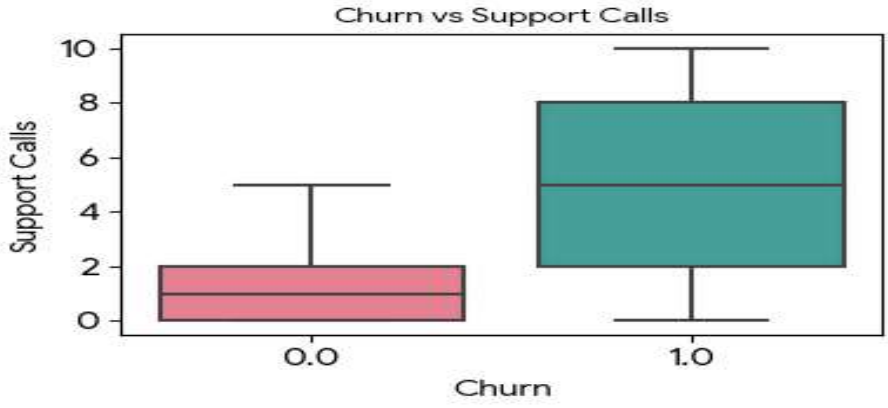

4.4Feature Correlation Heatmap :

A compact heatmap that identifies which metrics like `total spend or age` move together. It helps justify why certain features were selected for the model.

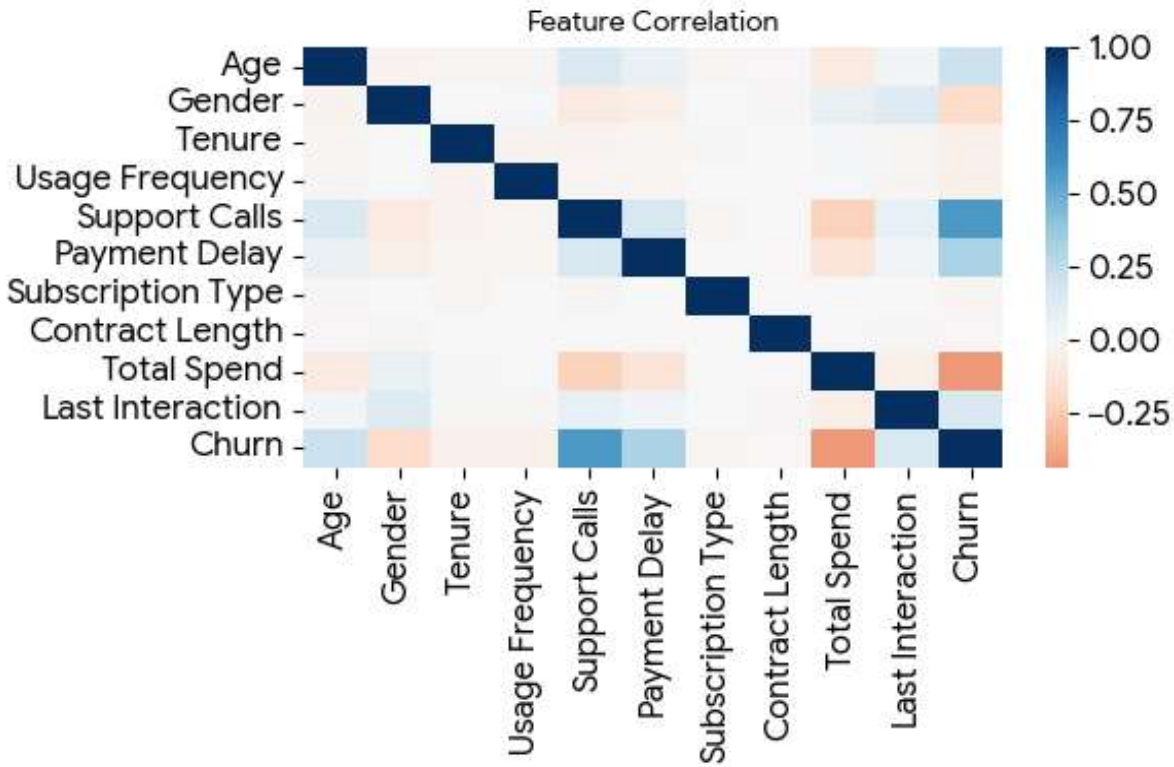

Figure 3 correlation heatmap

5. Results and Model Insights :

Random Forest Model Performance:

- **Accuracy:** 1.00
- **Precision (Churners / Non-churners):** 1.00 / 1.00
- **Recall (Churners / Non-churners):** 0.87 / 1.00
- **F1-score (Churners / Non-churners):** 0.93 / 1.00
- **ROC-AUC Score:** 1.00

- The model achieves **perfect overall accuracy and ROC-AUC**, indicating strong predictive capability.

- Most influential features include **customer tenure, activity levels, usage patterns, support interactions, and subscription type**.
- While overall performance is excellent, the **recall for churners (0.87)** indicates some churners are still being missed. Businesses can focus on this group for **targeted retention strategies**.

6. Conclusion :

The Random Forest model effectively predicts customer churn with high accuracy and ROC-AUC.
Key drivers of churn include customer tenure, activity levels, usage patterns, support interactions, and subscription type.
Exploratory Data Analysis and feature importance confirm that low engagement and shorter tenure are strongly associated with churn.
Although the model performs well overall, the recall for churners indicates some at-risk customers may still be missed.
Targeted retention strategies for these customers can help reduce churn and improve business performance.
These insights provide actionable guidance for enhancing customer engagement and driving better business outcomes.